

Password Strength Checker System

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main() {
    string password;
    int score = 0;

    cout << "Enter your password: ";
    cin >> password;

    // Length check
    if (password.length() >= 8)
        score += 20;
    if (password.length() >= 12)
        score += 10;

    bool hasLower = false, hasUpper = false, hasDigit = false, hasSpecial = false;

    for (char ch : password) {
        if (islower(ch))
            hasLower = true;
        else if (isupper(ch))
            hasUpper = true;
        else if (isdigit(ch))
            hasDigit = true;
        else
            hasSpecial = true;
    }

    // Add score
    if (hasLower) score += 15;
    if (hasUpper) score += 15;
    if (hasDigit) score += 15;
    if (hasSpecial) score += 15;

    // Common passwords check
    string common[] = {"123456", "password", "qwerty", "abc123"};
    for (string cp : common) {
        if (password == cp) {
            score -= 20;
            break;
        }
    }

    // Limit score between 0 and 100
    if (score < 0) score = 0;
    if (score > 100) score = 100;

    // Strength level
    string strength;
    if (score <= 30)
        strength = "Weak";
    else if (score <= 60)
        strength = "Moderate";
    else if (score <= 80)
        strength = "Strong";
    else
        strength = "Very Strong";

    // Output
    cout << "\nPassword Strength: " << score << "%" << endl;
    cout << "Strength Level: " << strength << endl;

    return 0;
}
```